

COTONOU, Benin

WMO: 653440

Lat: 6.357N

Lon: 2.384E

Elev: 6

StdP: 101.26

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.0	22.8	12.8	9.2	28.6	16.8	12.0	26.7	9.1	26.9	8.4	26.9	2.5	20	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	5.3	33.0	26.9	32.4	27.0	32.1	27.1	28.9	31.5	28.4	31.0	28.1	30.7	5.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	24.3	30.9	27.8	23.9	30.7	27.2	23.0	30.0	94.0	32.0	92.2	31.4	90.5	31.2	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.4	7.0	DB	20.2	34.5	1.7	1.1	19.0	35.3	18.0	36.0	17.0	36.6	15.8	37.5	(4)
(5)			WB	16.0	29.4	2.4	1.1	14.3	30.2	12.9	30.9	11.6	31.5	9.9	32.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	28.1	29.2	29.4	29.0	28.3	27.0	26.3	26.0	26.5	27.2	28.4	28.4	(6)	
	DBStd	1.53	1.12	1.05	1.19	1.34	1.31	1.12	0.85	0.71	0.81	0.99	0.84	0.90	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6500	560	536	602	570	568	510	506	495	496	534	552	571	(10)	
	CDD18.3	3458	302	303	344	320	309	260	247	237	246	276	302	313	(11)	
	CDH23.3	37667	3358	3770	4427	4034	3713	2597	2011	1651	2093	2782	3573	3658	(12)	
	CDH26.7	12962	1213	1626	2037	1784	1467	636	244	159	319	731	1339	1407	(13)	
(14) Wind	WSAvg	4.0	3.3	4.2	4.6	4.2	3.5	3.8	4.9	5.1	4.5	3.5	3.3	3.2	(14)	
(15) Precipitation	PrecAvg	955	13	26	47	109	171	245	95	22	103	108	23	7	(15)	
	PrecMax	1458	67	119	116	322	404	445	449	110	383	269	76	42	(16)	
	PrecMin	544	0	0	16	7	74	132	21	2	11	6	1	0	(17)	
	PrecStd	259	19	32	24	75	82	101	95	28	85	71	21	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	33.2	33.2	33.2	33.0	31.3	29.8	29.2	30.1	31.2	32.5	33.1	(19)	
		MCWB	22.2	26.3	27.5	28.0	27.7	26.6	25.9	25.2	26.0	26.7	26.4	24.4	(20)	
	2%	DB	32.0	32.4	32.9	32.8	32.2	30.8	29.1	28.6	29.3	31.0	32.1	32.2	(21)	
		MCWB	25.5	27.1	27.8	27.9	27.4	26.6	25.4	24.9	25.6	26.7	26.9	26.0	(22)	
	5%	DB	31.2	32.1	32.2	32.1	31.9	30.0	28.6	28.1	29.0	30.2	31.7	31.9	(23)	
		MCWB	26.0	27.2	27.5	27.6	27.2	26.3	25.2	24.7	25.4	26.3	27.0	26.3	(24)	
	10%	DB	30.9	31.7	31.9	31.8	31.1	29.3	28.0	27.8	28.4	29.9	31.1	31.1	(25)	
		MCWB	26.1	27.2	27.4	27.5	27.0	26.0	25.0	24.6	25.2	26.2	26.9	26.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.7	29.2	29.2	29.2	28.7	27.7	26.6	26.2	26.7	27.7	28.4	28.7	(27)	
		MCDB	30.8	31.5	32.1	31.9	31.4	29.7	28.4	27.8	28.6	29.9	31.0	31.0	(28)	
	2%	WB	28.0	28.7	28.8	28.7	28.1	27.1	26.1	25.7	26.2	27.1	28.0	28.1	(29)	
		MCDB	30.2	31.1	31.7	31.5	31.0	29.4	28.0	27.4	28.3	29.8	30.7	30.6	(30)	
	5%	WB	27.5	28.1	28.2	28.1	27.6	26.6	25.8	25.2	25.9	26.7	27.6	27.6	(31)	
		MCDB	29.7	30.5	30.9	30.8	30.6	29.0	27.6	27.1	28.1	29.3	30.4	30.1	(32)	
	10%	WB	27.1	27.7	27.8	27.7	27.2	26.2	25.4	25.0	25.6	26.4	27.1	27.1	(33)	
		MCDB	29.4	30.1	30.5	30.5	30.1	28.5	27.3	26.9	27.7	28.8	30.0	29.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	5.5	5.3	5.5	5.6	4.7	3.8	3.8	4.3	5.2	5.9	6.5	(35)	
		MCDBR	6.9	5.7	5.3	5.4	5.7	5.4	4.2	4.2	4.8	5.7	6.3	7.1	(36)	
	5% WB	MCWBR	3.4	2.8	2.8	2.8	2.8	2.7	2.1	2.0	2.3	2.8	2.8	3.1	(37)	
		MCWBR	5.6	5.2	5.2	5.3	5.6	5.1	4.0	4.0	4.6	5.5	5.9	5.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.834	0.840	0.827	0.662	0.555	0.524	0.529	0.532	0.489	0.509	0.603	0.708	(40)	
		taud	1.499	1.500	1.515	1.808	2.042	2.146	2.128	2.110	2.262	2.196	1.899	1.694	(41)	
	Ebn at Noon	Ebn at Noon	570	585	600	697	757	771	771	783	831	815	728	643	(42)	
		Edh at Noon	298	305	302	222	171	151	155	162	141	150	198	242	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	4.97	4.96	5.19	5.24	4.75	3.95	4.17	4.29	4.46	4.84	5.11	4.96	(44)	
	RadStd	RadStd	0.17	0.26	0.27	0.22	0.24	0.27	0.33	0.28	0.22	0.15	0.14	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.28	+0.31	N/A	+0.36	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.28	+0.31	N/A	+0.36	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page